

Today 12:09 PM

I am a mathematician with a background in PDEs, mathematical physics and probability.

I'm particularly interested in understanding the interplay between contraction and unitary semigroups generated by self-adjoint second-order elliptic operators. I have some exposure to the two types of semigroups, but I feel that they are almost always treated separately in the literature (and I never understood why):

On the contraction side, there's usually an L^2 and a C_0 theory. They agree on their overlap, and they both end up providing weak solutions for parabolic PDEs (with distributional data in L^2 and C_0). In the L^2 picture, there are two usual approaches: global analysis (Galerkin method, spectral methods, maximum principle) and microlocal analysis (symbolic inversion, parametrices, finer regularity control). The C_0 theory tends to be more dynamical and probabilistic (Markov semigroups, Feynman-Kac equation).

On the unitary side, representation theory and microlocal analysis provide powerful tools. Unitary semigroups are often understood as the (Weyl) quantization of Hamiltonian flows, with Schwartz kernel phases "remembering" the classical Hamiltonian evolution. In this framework, both classical and quantum systems can be studied by how observables evolve, with Egorov's theorem establishing a bridge between the classical and the quantum setting. Moreover, representation theory

makes the choice of functional space here almost canonical: one can always take the Schrodinger L^2 space, thanks to the Stone-Von Neumann theorem.

With your help, I'd like to connect different notions under a more unified framework. As I mentioned above, a common theme seems to be dynamical representations of these semigroups. I know some cases like operators of "hamonic oscillator" type (i.e quantization of quadratic Hamiltonians have a very explicit bridge)

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A useful starting point is to treat both evolutions as boundary values of one holomorphic functional calculus, then ask what additional structure survives on each boundary. I'll organize the bridge at the abstract, PDE, probabilistic, and microlocal levels.

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Your instinct is right: the separation is largely **methodological rather than structural**. For a nonnegative self-adjoint elliptic operator, the contraction and unitary evolutions are two boundary regimes of a single complex-time object. What changes dramatically between the regimes is not the spectral theory, but the additional structures one can exploit: positivity and probability on one side, symplectic propagation and oscillation on the other.

I will write $A \geq 0$ for a self-adjoint realization of a second-order elliptic operator on

$$\mathcal{H} = L^2(M, E; d\mu).$$

1. The master object is the complex-time semigroup

Define

$$S(z) = e^{-zA} = \int_{[0, \infty)} e^{-z\lambda} dE_A(\lambda), \quad \Re z > 0.$$

Then:

$$S(t) = e^{-tA}, \quad t > 0$$

is the heat/contraction semigroup, while

$$S(it) = e^{-itA}, \quad t \in \mathbb{R}$$

is the Schrödinger unitary group.

For every $f \in \mathcal{H}$,

$$\lim_{\varepsilon \downarrow 0} S(\varepsilon + it)f = e^{-itA}f$$

strongly, by dominated convergence in the spectral representation. Thus the Wick rotation is completely rigorous at the L^2 operator level. The spectral-calculus construction of e^{-zA} for $\Re z > 0$ is precisely the natural common framework.  ar...

It is important that this convergence is generally **not in operator norm**. When A is unbounded,

$$\left\| e^{-(\varepsilon+it)A} - e^{-itA} \right\| = \sup_{\lambda \in \sigma(A)} |e^{-\varepsilon\lambda} - 1| = 1.$$

So the imaginary axis is a genuinely singular boundary of the bounded holomorphic semigroup, even though strong boundary values exist.

For the free Laplacian this is visible directly:

$$K_z(x, y) = (4\pi z)^{-n/2} \exp\left(-\frac{|x-y|^2}{4z}\right), \quad \Re z > 0.$$

As $z \rightarrow it$, this becomes distributionally

$$K_{it}(x, y) = (4\pi it)^{-n/2} \exp\left(\frac{i|x-y|^2}{4t}\right).$$

The same complex Gaussian changes from exponentially decaying to oscillatory.

This is the simplest prototype for the entire story.

2. Why the two theories nevertheless look different

A useful slogan is:

Heat evolution sees order and probability; Schrödinger evolution sees phase space

For e^{-tA} , the multiplier $e^{-t\lambda}$ suppresses high frequencies. Consequently, for elliptic A , one obtains smoothing, positive kernels in scalar situations, maximum

principles, L^p -contractivity, Dirichlet-form methods and often a Feller action on $C_0(M)$.

For e^{-itA} , the multiplier has modulus one. Singularities are not destroyed but transported. The natural questions become propagation of wavefront sets, caustics, dispersive estimates and Egorov-type conjugation of observables.

This also explains the functional-space asymmetry. Heat kernels are often integrable and positivity preserving, so L^p and C_0 extensions arise naturally. Schrödinger propagators are oscillatory and generally do not act boundedly on L^p , $p \neq 2$, without derivative loss or additional localization. There is a substantial theory deriving L^p -Sobolev estimates for e^{itA} from heat-kernel hypotheses, but the estimates themselves demonstrate that L^2 is exceptional.

✖ ar...

On the probabilistic side, the distinction is even sharper. Feynman–Kac expresses e^{-tA} using an actual positive Wiener measure:

$$e^{-t(-\frac{1}{2}\Delta+V)}f(x) = \mathbb{E}_x \left[e^{-\int_0^t V(B_s) ds} f(B_t) \right].$$

✖ ar...

For the real-time Schrödinger propagator, there is generally no analogous countably additive complex measure on continuous path space having the formal Feynman finite-dimensional distributions. Rigorous real-time constructions therefore use oscillatory integrals, time slicing, white-noise distributions or other generalized notions rather than ordinary probability measures.

✖ ar...

So “analytic continuation of formulas” is valid, but “analytic continuation of probability measures” usually is not.

3. The wave group provides another exact bridge

Although the direct unitary partner of e^{-tA} is e^{-itA} , the unitary evolution that most clearly remembers the metric geometry is

$$e^{is\sqrt{A}} \quad \text{or} \quad \cos(s\sqrt{A}).$$

The spectral theorem gives the transmutation identity

$$e^{-tA} = \frac{1}{\sqrt{4\pi t}} \int_{\mathbb{R}} e^{-s^2/(4t)} \cos(s\sqrt{A}) ds.$$

Thus the heat flow is a Gaussian average of wave propagation.

This is more than a formal relation. For nonnegative self-adjoint operators on metric measure spaces, finite propagation speed for $\cos(s\sqrt{A})$ is equivalent to Davies–Gaffney off-diagonal estimates for e^{-tA} . In this sense, Gaussian heat localization is the averaged remnant of sharp hyperbolic propagation.  ar...

Schematically,

$$\text{finite propagation of waves} \iff \text{off-diagonal heat decay}.$$

This is one of the cleanest points where hyperbolic, parabolic and spectral theories become literally equivalent.

4. Ground-state transforms connect quantum Hamiltonians to Markov generators

Suppose

$$A\phi_0 = E_0\phi_0, \quad \phi_0 > 0.$$

Define

$$U : L^2(M, \phi_0^2 d\mu) \longrightarrow L^2(M, d\mu), \quad Uf = \phi_0 f,$$

and

$$L = U^{-1}(A - E_0)U.$$

For $A = -\Delta + V$ on $\mathbb{R}^n$,

$$L = -\Delta - 2\nabla \log \phi_0 \cdot \nabla.$$

In particular,

$$L1 = 0,$$

and, under the standard locality assumptions, $-L$ is a diffusion generator symmetric with respect to $\phi_0^2 dx$. A Schrödinger operator with a positive ground state is thereby transformed into a Dirichlet-form operator and a Markov semigroup.  ar...

The harmonic oscillator is the canonical example:

$$A = -\Delta + |x|^2, \quad E_0 = n, \quad \phi_0(x) = e^{-|x|^2/2}.$$

Then

$$L = \phi_0^{-1}(A - n)\phi_0 = -\Delta + 2x \cdot \nabla$$

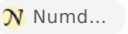
on $L^2(e^{-|x|^2} dx)$. Consequently,

$$-L = \Delta - 2x \cdot \nabla$$

is the Ornstein–Uhlenbeck generator.

Thus:

- oscillator heat evolution is conjugate to the Ornstein–Uhlenbeck Markov semigroup;
- oscillator real-time evolution is conjugate to the unitary group generated by the same self-adjoint Ornstein–Uhlenbeck operator;
- the Mehler formula is simultaneously a quantum, semigroup and probabilistic formula.

There is even a Gaussian Weyl calculus specifically adapted to the Ornstein–Uhlenbeck operator, including the complex-time operators e^{-zL} . 

This ground-state/Doob transform is probably the most useful bridge beyond exactly quadratic Hamiltonians.

5. Microlocally, heat evolution uses complexified Hamiltonian geometry

Let

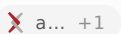
$$A_h = \text{Op}_h(p) + \text{lower-order terms},$$

with p real and elliptic. The semiclassical Schrödinger propagator

$$e^{-itA_h/h}$$

is, locally in time and away from the usual degeneracies, a Fourier integral operator associated with

$$\text{graph}(\Phi_t), \quad \Phi_t = \exp(tH_p).$$

Quadratic Hamiltonians give exact Gaussian/FIO formulas rather than merely asymptotic parametrices. 

The corresponding complex-time family is

$$e^{-zA_h/h}, \quad z = \tau + it, \quad \tau \geq 0.$$

Formally, the underlying canonical transformation is

$$\exp((-iz)H_p) = \exp((t - i\tau)H_p),$$

up to the usual sign convention for propagators.

At $\tau = 0$, this is the real Hamiltonian flow. For $\tau > 0$, it moves into the complexified cotangent bundle, and the associated Lagrangian acquires a positive imaginary part. That positive complex geometry is the microlocal manifestation of damping.

There is one important regularity qualification:

- for analytic coefficients, complexified Hamiltonian flows and complex-phase FIOs can often be taken literally;
- for merely C^∞ coefficients, the Volterra heat calculus and pseudodifferential parametrices are generally more robust than an actual holomorphic complexified flow;
- for quadratic p , the complex symplectic picture is global and exact.

So the heat kernel does “remember” Hamiltonian dynamics, but usually through a **positive complex Lagrangian**, rather than a real canonical graph.

6. The oscillator semigroup is the representation-theoretic answer in the quadratic case

For real quadratic Hamiltonians, the unitary propagators form the metaplectic representation of the real symplectic group. Howe’s oscillator semigroup extends this representation into a complex symplectic semigroup.

Its interior consists of Gaussian contraction operators; its boundary contains the metaplectic representation. Hilgert describes this as a contractive representation of a covering semigroup whose Shilov boundary is the metaplectic group, with Gaussian integral kernels in the interior. Numd...

Thus the representation-theoretic picture is literally:

$$\begin{array}{ccc} \text{positive complex symplectic maps} & \longmapsto & \text{Gaussian contractions} \\ \partial & & \partial \\ \text{real symplectic maps} & \longmapsto & \text{metaplectic unitaries.} \end{array}$$

A recent systematic treatment formulates this as a metaplectic semigroup associated with positive complex symplectic matrices and applies it to quadratic parabolic equations, evolution operators and propagation of Wigner singularities. ar...

This is very likely the precise formalism underlying the explicit “harmonic-oscillator bridge” you have in mind.

7. Contraction semigroups are also compressions of unitary dynamics

There is a separate, completely abstract unification. A strongly continuous contraction semigroup T_t on a Hilbert space has a unitary dilation: on a larger Hilbert space $\mathcal{K} \supset \mathcal{H}$, one can find a unitary group U_t such that

$$T_t = P_{\mathcal{H}} U_t|_{\mathcal{H}}, \quad t \geq 0.$$

So dissipative evolution can always be regarded as conservative evolution with inaccessible degrees of freedom. Sz.-Nagy's dilation theory includes one-parameter contraction semigroups, and has a natural system-theoretic interpretation in terms of embedding a dissipative system into a conservative input/state/output system.  ar...

This framework is conceptually powerful, especially for observables:

$$B \longmapsto T_t^* B T_t$$

is completely positive and subunital, while its dilation is obtained from the automorphism

$$\tilde{B} \longmapsto U_t^* \tilde{B} U_t.$$

However, a generic unitary dilation usually forgets the locality, ellipticity and principal symbol of the original PDE. It is an operator-theoretic unification, not automatically a microlocal one.

8. A qualification concerning Stone-von Neumann

Stone-von Neumann makes the Schrödinger representation canonical only for **finite-dimensional, regular, irreducible representations of the Weyl CCR**. It does not by itself canonically select $L^2(M)$ for an arbitrary configuration manifold or elliptic operator. In infinite-dimensional CCR systems, unitarily inequivalent representations occur abundantly.  ar...

For an elliptic operator on a manifold, the natural Hilbert space is more properly determined by the configuration geometry, density or half-density, bundle data and self-adjoint realization. Stone-von Neumann explains the special rigidity of $T^*\mathbb{R}^n$, not a universal uniqueness of quantum Hilbert spaces.

A workable unified programme

A particularly coherent object to study is

$$S_h(z) = e^{-zA_h/h}, \quad \Re z \geq 0.$$

The main questions then become:

1. **Spectral:** boundary values, resolvents, fractional powers and spectral multipliers.
2. **Microlocal:** uniform kernel parametrices as $\Re z \downarrow 0$, and the degeneration of positive complex Lagrangians into real canonical graphs.
3. **Probabilistic:** Feynman–Kac and Dirichlet-form representations in the interior of the half-plane.
4. **Representation-theoretic:** oscillator/metaplectic semigroups in the quadratic approximation.
5. **Dynamical:** unitary dilation and the evolution of observables.
6. **Geometric:** ground-state transforms converting Schrödinger operators into symmetric diffusion generators.

The harmonic oscillator sits at the intersection of all six viewpoints. The natural next model is a confining Schrödinger operator

$$-\Delta + V(x)$$

with a positive ground state: the Doob transform retains the probabilistic structure, while semiclassical complex-time analysis retains the Hamiltonian structure. This provides a nonquadratic setting in which the two languages can genuinely interact rather than merely coexist.

Today 2:14 PM

Michael Taylor - Par...
PDF

Indeed, I agree that the quantum harmonic oscillator is a central example, and I'd like to understand it in depth.

I'm sending you a textbook reference that contains (in a scattered form) most of the key points you highlighted for the quantum harmonic oscillator. I'd like you to read it carefully and use it as a base reference to give a unified account of the following narrative:

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Taylor's harmonic oscillator as a meeting point

The most productive way to read Taylor's treatment is **transversally across Chapters 7, 8, and 11**. He does not intend to give a unified account of quantum mechanics, Wick rotation, or the metaplectic representation. In fact, the preface explicitly says that the book gives relatively little attention to quantum mechanics and does not discuss the physical theories behind the quantum-mechanical operators it treats.

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Instead, the oscillator is reused for different methodological purposes:

- Chapter 7: a model for the Weyl calculus, global ellipticity, heat parametrices, and oscillator-adapted Sobolev spaces;
- Chapter 8: an exactly solvable self-adjoint operator, illustrating spectral algebra and Lie-theoretic factorization;
- Chapter 11: a quadratic potential to which Feynman-Kac and stochastic methods can be applied.

Taylor explicitly presents the Weyl calculus as a symmetry-sensitive alternative that produces a more efficient heat parametrix, and then introduces the oscillator class because it treats x and ξ symmetrically.

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In Chapter 8, the goal is an explicit formula for e^{-tH} , partly because analogues of Mehler's formula will later be used in the heat-equation proof of the index theorem.

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Chapter 11 then reorganizes heat evolution probabilistically through Wiener measure, Feynman-Kac, stochastic integrals, and SDEs.

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What follows is therefore partly an account of Taylor and partly a synthesis of the structures that Taylor leaves implicit.

1. Normalization and the central operator

Taylor uses

$$H = -\Delta + |x|^2 = \text{Op}^w(q), \quad q(x, \xi) = |x|^2 + |\xi|^2$$

on $L^2(\mathbb{R}^n, dx)$.

His Brownian motion convention is adapted to the heat semigroup $e^{t\Delta}$, rather than $e^{t\Delta/2}$. Thus

$$\mathbb{E}[(\omega_t - \omega_s)^2] = 2|t - s|$$

in one dimension. Taylor explicitly warns that this convention changes the factors $1/2$ appearing in the usual probabilistic formulas.

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The two formulas that will eventually organize everything are

$$e^{-zH} = \text{Op}^w\left((\cosh z)^{-n} e^{-(\tanh z)(|x|^2 + |\xi|^2)}\right), \quad \Re z > 0,$$

and, after the ground-state transformation,

$$U^{-1} e^{-zH} U = e^{-nz} e^{-z\mathcal{L}}, \quad \mathcal{L} = -\Delta + 2x \cdot \nabla.$$

The first is the phase-space or symplectic description; the second is the Gaussian/probabilistic description.

2. The generic parametrix and why the oscillator is exceptional

2.1 Taylor's ordinary heat parametrix

Chapter 7, §13, book pp. 56–67 (PDF pp. 79–90), treats a general second-order elliptic operator L with positive scalar principal symbol $L_2(x, \xi)$. Taylor seeks an approximate heat solution in oscillatory-integral form

$$u(t, x) = \int e^{ix \cdot \xi} a(t, x, \xi) \hat{f}(\xi) d\xi, \quad a \sim \sum_{j \geq 0} a_j.$$

The leading amplitude is

$$a_0(t, x, \xi) = e^{-tL_2(x, \xi)},$$

and the a_j are obtained recursively from transport equations.

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After asymptotic summation, the resulting operator differs from the true heat semigroup by a smooth kernel that vanishes rapidly as $t \downarrow 0$.

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This construction is:

- local in x ;
- asymptotic as $t \downarrow 0$;
- organized by symbol order;

- insensitive, at first, to special global symmetries of a particular operator.

It gives the universal short-time Gaussian

$$(4\pi t)^{-n/2} \exp\left(-\frac{d(x, y)^2}{4t}\right)$$

with recursively determined corrections.

2.2 The Weyl improvement

In §14 Taylor replaces the left quantization by Weyl quantization:

$$\text{Op}^w(a)u(x) = (2\pi)^{-n} \iint a\left(\frac{x+y}{2}, \xi\right) e^{i(x-y)\cdot\xi} u(y) dy d\xi.$$

The Weyl product begins

$$a\#b = ab + \frac{i}{2}\{a, b\} + \dots.$$

The decisive observation is that if the approximate heat symbol is a function of the principal symbol,

$$b = f(a),$$

then

$$\{a, f(a)\} = 0.$$

The first correction in the Weyl product therefore vanishes. Taylor emphasizes that this makes the Weyl parametrix more accurate at a given stage than the ordinary pseudodifferential parametrix.

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There is also a deeper structural advantage. The Heisenberg representation makes the natural symplectic form visible, and linear symplectic transformations act by automorphisms of the Weyl product:

$$a\#b = c \implies (a \circ g^{-1})\#(b \circ g^{-1}) = c \circ g^{-1}, \quad g \in Sp(n, \mathbb{R}).$$

Taylor develops this through the Heisenberg group and formulas (14.14)–(14.19).

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2.3 Why the oscillator parametrix closes exactly

For

$$q(x, \xi) = |x|^2 + |\xi|^2,$$

the Weyl product $q \# h$ is not merely asymptotic. Because q is quadratic, all derivatives of order greater than two vanish, so the Moyal expansion terminates exactly.

Writing

$$e^{-tH} = \text{Op}^w(h_t),$$

Taylor obtains

$$\partial_t h_t = -qh_t + \frac{1}{4}\Delta_{x,\xi}h_t, \quad h_0 = 1.$$

Symplectic symmetry suggests the radial ansatz

$$h_t(x, \xi) = g(t, Q), \quad Q = |x|^2 + |\xi|^2.$$

It reduces the $2n$ -dimensional symbol equation to

$$\partial_t g = -Qg + Q\partial_Q^2 g + n\partial_Q g.$$

The Gaussian ansatz

$$g(t, Q) = a(t)e^{b(t)Q}$$

then gives

$$b' = b^2 - 1, \quad \frac{a'}{a} = nb, \quad b(0) = 0, \quad a(0) = 1.$$

Hence

$$b(t) = -\tanh t, \quad a(t) = (\cosh t)^{-n},$$

and therefore

$$h_t(x, \xi) = (\cosh t)^{-n} \exp [-(\tanh t)(|x|^2 + |\xi|^2)].$$

This is Taylor's formulas (15.32)–(15.42). He also explains why the radial ansatz is forced by the $U(n) \subset Sp(n, \mathbb{R})$ invariance of q .

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This is the first major unification:

The generic heat parametrix is an infinite asymptotic symbolic construction. For a quadratic Hamiltonian, the Weyl calculus truncates, symmetry reduces the equation, and the parametrix becomes the exact semigroup.

3. From the Weyl symbol to Mehler's kernel

Applying the Weyl kernel formula to h_t gives a Gaussian Fourier integral. Its evaluation yields

$$K_t^H(x, y) = \frac{1}{(2\pi \sinh 2t)^{n/2}} \exp \left[-\frac{\cosh 2t (|x|^2 + |y|^2) - 2x \cdot y}{2 \sinh 2t} \right].$$

Thus

$$e^{-tH} f(x) = \int_{\mathbb{R}^n} K_t^H(x, y) f(y) dy.$$

Taylor first obtains this formula through the Lie-algebra factorization discussed below, and then recovers the simpler Weyl symbol by an inverse Gaussian transform. See Chapter 8, §6, formulas (6.33)–(6.43).

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The two forms encode different geometry:

- the kernel form uses x, y and displays diffusion and off-diagonal decay;
- the Weyl-symbol form uses (x, ξ) and displays symplectic invariance.

The apparently different hyperbolic functions,

$$\tanh t, \quad \sinh 2t, \quad \cosh 2t,$$

are related by the Gaussian Fourier transform that passes between the Weyl symbol and the Schwartz kernel.

Taylor also extends the formula from the isotropic oscillator to a general positive quadratic form Q . By symplectic diagonalization and the Hamilton map F_Q , he obtains a coordinate-free formula involving

$$A_Q = (-F_Q^2)^{1/2}, \quad \frac{\tanh(tA_Q)}{tA_Q}.$$

This is formulas (6.47)–(6.55).

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4. Taylor's Lie-algebra derivation

4.1 The $\mathfrak{sl}_2$ triple

In dimension one Taylor introduces

$$X = -\partial_x^2, \quad Y = x^2, \quad Z = x\partial_x + \partial_x x = 2x\partial_x + 1.$$

Then $H = X + Y$, and

$$[X, Y] = -2Z, \quad [X, Z] = 4X, \quad [Y, Z] = -4Y.$$

These span a copy of

$$\mathfrak{sl}_2(\mathbb{R}) \cong \mathfrak{sp}(1, \mathbb{R}).$$

Taylor identifies this triple with the standard upper-triangular, lower-triangular, and diagonal generators of $\mathfrak{sl}_2$, then performs a Gauss decomposition of

$$e^{-2t(n_+ + n_-)}.$$

The resulting scalar parameters are

$$\varepsilon_1(t) = \varepsilon_2(t) = \frac{1}{2} \tanh 2t, \quad e^{2\varepsilon_3(t)} = \cosh 2t.$$

This motivates the operator factorization

$$e^{-tH} = e^{-\varepsilon_1(t)X} e^{\varepsilon_3(t)Z} e^{-\varepsilon_2(t)Y}.$$

The three factors are elementary:

$$e^{-\varepsilon X} = e^{\varepsilon \partial_x^2},$$

the free heat semigroup;

$$e^{-\varepsilon Y} f(x) = e^{-\varepsilon x^2} f(x),$$

Gaussian killing;
and

$$e^{sZ} f(x) = e^s f(e^{2s} x),$$

a unitary dilation on $L^2(\mathbb{R})$.

Composing these three kernels gives Mehler's formula. Taylor's complete algebraic calculation is in formulas (6.21)–(6.34).

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4.2 What Taylor deliberately does not prove

Taylor explicitly says that the Lie-algebra homomorphism should integrate to a representation of a covering group of $SL(2, \mathbb{R})$, but that there are analytic issues—particularly because some of the relevant operators are unbounded—and he therefore verifies the kernel directly instead.

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The standard representation-theoretic completion of his argument is:

- the covering group is the metaplectic group;
- real symplectic matrices act through metaplectic unitary operators;
- appropriate positive complex symplectic matrices act through Gaussian contractions, forming the oscillator or metaplectic semigroup.

Taylor's heat factorization is the interior, contractive version of a metaplectic factorization.

5. The unitary oscillator and exact Egorov dynamics

This is the first major step beyond Taylor's explicit presentation.

The classical Hamiltonian

$$q(x, \xi) = |x|^2 + |\xi|^2$$

has Hamilton equations

$$\dot{x} = 2\xi, \quad \dot{\xi} = -2x.$$

Thus

$$\Phi_t = \begin{pmatrix} \cos 2t & \sin 2t \\ -\sin 2t & \cos 2t \end{pmatrix}.$$

The unitary group

$$U_t = e^{-itH}$$

is the metaplectic lift of Φ_t . Consequently, for every suitable symbol a ,

$$\boxed{U_t^* \operatorname{Op}^w(a) U_t = \operatorname{Op}^w(a \circ \Phi_t).}$$

For quadratic dynamics, Egorov's theorem is exact: there are no lower-order or semiclassical error terms.

Taylor's symplectic covariance formula (14.19) already establishes that the transformation $a \mapsto a \circ g^{-1}$ preserves the Weyl product. What is missing there is only the explicit construction of the metaplectic implementer.

6. Analytic continuation to imaginary time

For $\Re z > 0$, the spectral theorem defines a holomorphic family

$$e^{-zH}.$$

Taylor's formulas analytically continue to

$$h_z(x, \xi) = (\cosh z)^{-n} \exp [-(\tanh z)q(x, \xi)].$$

For $z = it$, initially on an interval where $\cos t \neq 0$,

$$h_{it}(x, \xi) = (\cos t)^{-n} \exp [-i(\tan t)(|x|^2 + |\xi|^2)].$$

This is the Weyl symbol, understood with the branch obtained as the boundary value

$$h_{it} = \lim_{\varepsilon \downarrow 0} h_{\varepsilon + it}.$$

Similarly, Mehler's heat kernel continues to

$$K_{it}^H(x, y) = \frac{1}{(2\pi i \sin 2t)^{n/2}} \exp \left[i \frac{\cos 2t (|x|^2 + |y|^2) - 2x \cdot y}{2 \sin 2t} \right],$$

for $\sin 2t \neq 0$, again with the boundary-value branch.

The phase

$$S_t(x, y) = \frac{\cos 2t (|x|^2 + |y|^2) - 2x \cdot y}{2 \sin 2t}$$

is a generating function for the classical flow:

$$\xi = \partial_x S_t(x, y), \quad \eta = -\partial_y S_t(x, y)$$

implies

$$(x, \xi) = \Phi_t(y, \eta).$$

Thus the unitary kernel literally remembers the classical linear symplectic motion.

At the caustic times

$$t \in \frac{\pi}{2} \mathbb{Z},$$

the displayed kernel ceases to be an ordinary function, but the unitary group remains well defined distributionally. For example,

$$e^{-i\frac{\pi}{2}H} = e^{-i\pi n/2} \mathcal{P}, \quad (\mathcal{P}f)(x) = f(-x).$$

Conceptually, complex time replaces

$$\Phi_t \quad \text{by} \quad \Phi_{-iz}.$$

For positive real $z = t$,

$$\Phi_{-it} = \begin{pmatrix} \cosh 2t & -i \sinh 2t \\ i \sinh 2t & \cosh 2t \end{pmatrix},$$

a positive complex symplectic transformation. The hyperbolic functions in the heat kernel are therefore the imaginary-time versions of the trigonometric functions in the classical flow.

Taylor does not carry out this oscillator Wick rotation in §8.6; it is the natural continuation of the formulas he derives.

7. The ground state and the Ornstein-Uhlenbeck transform

Taylor's ladder-operator argument identifies the ground state and the entire spectrum:

$$\phi_0(x) = \pi^{-n/4} e^{-|x|^2/2}, \quad H\phi_0 = n\phi_0,$$

and

$$\text{spec } H = \{2k + n : k \geq 0\}.$$

The eigenfunctions are Hermite polynomials times ϕ_0 .

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Let

$$d\gamma(x) = \phi_0(x)^2 dx = \pi^{-n/2} e^{-|x|^2} dx,$$

and define the unitary map

$$U : L^2(\gamma) \longrightarrow L^2(dx), \quad Uf = \phi_0 f.$$

A direct calculation gives

$$H(\phi_0 f) = \phi_0 (-\Delta f + 2x \cdot \nabla f + nf).$$

Hence

$$U^{-1}(H - n)U = \mathcal{L} := -\Delta + 2x \cdot \nabla.$$

The negative operator

$$G = -\mathcal{L} = \Delta - 2x \cdot \nabla$$

is the Ornstein-Uhlenbeck Markov generator. It is symmetric with respect to γ :

$$\langle \mathcal{L}f, g \rangle_{L^2(\gamma)} = \int_{\mathbb{R}^n} \nabla f \cdot \nabla \bar{g} d\gamma.$$

Therefore

$$e^{-tH} = e^{-nt} U e^{-t\mathcal{L}} U^{-1} = e^{-nt} U P_t^{OU} U^{-1}.$$

The corresponding OU process satisfies, in Taylor's Brownian normalization,

$$dX_t = -2X_t dt + d\omega_t,$$

so

$$X_t = e^{-2t}x + \int_0^t e^{-2(t-s)} d\omega_s.$$

Consequently,

$$P_t^{OU} f(x) = \int_{\mathbb{R}^n} f\left(e^{-2t}x + \sqrt{1 - e^{-4t}} y\right) d\gamma(y).$$

Taylor develops the Langevin equation and its exponentially damped Gaussian solution in Chapter 11, §7, and explicitly points toward the work of Ornstein and Uhlenbeck. Michael Taylor - Partial Differ... His later SDE theory constructs Markov semigroups and identifies their second-order diffusion generators.

Michael Taylor - Partial Differ... What Taylor does not explicitly write is the conjugation connecting this process to H .

The Mehler kernel and the OU transition kernel are therefore the same Gaussian formula in two measures. If

$$p_t^{OU}(x, y) = \frac{\exp\left[-|y - e^{-2t}x|^2 / (1 - e^{-4t})\right]}{[\pi(1 - e^{-4t})]^{n/2}}$$

is the OU kernel relative to Lebesgue measure, then

$$K_t^H(x, y) = e^{-nt} \frac{\phi_0(x)}{\phi_0(y)} p_t^{OU}(x, y).$$

The ground-state transformation converts:

- the lowest energy n into the Markov eigenvalue 0;
- the ground state ϕ_0 into the constant function 1;
- Hermite functions into Hermite polynomials;
- the confining potential into an inward linear drift.

8. Feynman-Kac and the oscillator

Taylor derives Feynman-Kac using the Trotter product formula:

$$e^{t(\Delta-V)} = \lim_{N \rightarrow \infty} \left(e^{(t/N)\Delta} e^{-(t/N)V} \right)^N.$$

The finite products become finite-dimensional integrals against Gaussian transition densities. In the limit they become Wiener integrals:

$$e^{t(\Delta-V)} f(x) = \mathbb{E}_x \left[f(\omega_t) \exp \left(- \int_0^t V(\omega_s) ds \right) \right].$$

Taylor first proves this for bounded continuous V . Michael Taylor - Partial Differ... He then extends it to locally integrable potentials bounded below, which includes $V(x) = |x|^2$. Michael Taylor - Partial Differ...

For the oscillator,

$$e^{-tH} f(x) = \mathbb{E}_x \left[f(\omega_t) \exp \left(- \int_0^t |\omega_s|^2 ds \right) \right].$$

Taylor makes the Mehler-Feynman-Kac link especially explicit in the exercises following §2: he asks the reader to calculate

$$e^{t(\partial_x^2 - \varepsilon x^2 - \eta x)}$$

using Mehler's formula and then to use it to evaluate Brownian expectations involving

$$\exp \left(-\varepsilon \int_0^t \omega_s^2 ds - \eta \int_0^t \omega_s ds \right).$$

Thus the same formula is being obtained in two ways:

Lie/Weyl calculation

Trotter time slicing

↓

↓

Mehler Gaussian

quadratically weighted Wiener measure.

9. Feynman-Kac becomes Ornstein-Uhlenbeck after the ground-state transform

This is perhaps the cleanest missing link in Taylor.

Define

$$M_t = e^{nt} \frac{\phi_0(\omega_t)}{\phi_0(x)} \exp \left(- \int_0^t |\omega_s|^2 ds \right).$$

Then

$$P_t^{OU} f(x) = e^{nt} \phi_0(x)^{-1} e^{-tH} (\phi_0 f)(x) = \mathbb{E}_x[f(\omega_t) M_t].$$

Now use Taylor's Itô convention. Since

$$\log \phi_0(x) = -\frac{1}{2}|x|^2 + \text{constant},$$

Itô's formula gives

$$\log \phi_0(\omega_t) - \log \phi_0(x) = - \int_0^t \omega_s \cdot d\omega_s - nt.$$

Therefore

$$M_t = \exp \left[- \int_0^t \omega_s \cdot d\omega_s - \int_0^t |\omega_s|^2 ds \right].$$

But this is precisely the Girsanov exponential corresponding, in Taylor's normalization, to the drift

$$b(x) = -2x.$$

Taylor derives the general stochastic representation

$$e^{t(\Delta+X)} f(x) = \mathbb{E}_x \left[f(\omega_t) \exp \left(\frac{1}{2} \int_0^t X(\omega_s) \cdot d\omega_s - \frac{1}{4} \int_0^t |X(\omega_s)|^2 ds \right) \right]$$

for real vector fields X , initially under boundedness hypotheses.

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Taking $X(x) = -2x$ gives exactly the density above.

Taylor's Itô identity, including the quadratic-variation correction, is developed immediately afterward.

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The linear drift can be justified by localization or directly through his subsequent SDE theory, which permits coefficients of linear growth.

So there are two equivalent path-space descriptions:

$\text{Brownian motion killed at rate } x ^2 \xrightarrow{\text{ground-state/Doob transform}} \text{conservative}$

The potential has become a drift. This is the probabilistic form of

$$\phi_0^{-1}(H - n)\phi_0 = -\Delta + 2x \cdot \nabla.$$

10. Euclidean and real-time path integrals

10.1 Euclidean time

Taylor's proof of Feynman-Kac is already a rigorous Euclidean path integral. The Trotter product

$$\left(e^{\varepsilon \Delta} e^{-\varepsilon |x|^2} \right)^N, \quad \varepsilon = t/N,$$

has a kernel built from the discrete action

$$\sum_j \left[\frac{|x_{j+1} - x_j|^2}{4\varepsilon} + \varepsilon |x_j|^2 \right].$$

The kinetic term becomes Wiener measure; the potential term remains the multiplicative functional

$$e^{-\int_0^t |\omega_s|^2 ds}.$$

Taylor's Appendix A isolates the abstract Trotter theorem underlying this construction.

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Equivalently, after conditioning on the endpoint, the heat kernel can be written as a Brownian-bridge expectation:

$$K_t^H(x, y) = p_t(x, y) \mathbb{E}_{x \rightarrow y}^t \left[e^{-\int_0^t |\omega_s|^2 ds} \right].$$

10.2 Real time

For the unitary group, the corresponding Trotter formula is

$$e^{-itH} = \text{s-lim}_{N \rightarrow \infty} \left(e^{i(t/N)\Delta} e^{-i(t/N)|x|^2} \right)^N.$$

The free kernel is now

$$(4\pi i \varepsilon)^{-n/2} \exp \left(\frac{i|x-y|^2}{4\varepsilon} \right),$$

so the finite-dimensional integrals involve

$$\exp \left\{ i \sum_j \left[\frac{|x_{j+1} - x_j|^2}{4\varepsilon} - \varepsilon |x_j|^2 \right] \right\}.$$

Formally, the continuum action is

$$S[x] = \int_0^t \left(\frac{1}{4} |\dot{x}(s)|^2 - |x(s)|^2 \right) ds,$$

with the factor $1/4$ resulting from Taylor's normalization $H = -\Delta + |x|^2$.

For the oscillator, all finite-dimensional integrals are Fresnel-Gaussian integrals. They can be evaluated exactly, and the limit is the oscillatory Mehler kernel above.

The important distinction is:

Wick rotation analytically continues the kernels and finite-dimensional integrals

but it does not turn Wiener measure into an ordinary countably additive “complex probability measure.” The real-time path integral is naturally an oscillatory integral or a limit of time-sliced Fresnel integrals, not a stochastic expectation in the usual probabilistic sense.

Taylor supplies the rigorous Euclidean half of this statement. The real-time continuation is the representation-theoretic completion.

11. The common finite-dimensional mechanism

All the approaches close because the oscillator is quadratic.

Symbolic closure

The Moyal product with a quadratic symbol terminates after the second-order term. The infinite symbolic transport hierarchy becomes one scalar Riccati equation.

Symplectic closure

The Hamiltonian flow is linear. Its quantization lies in the metaplectic representation, so the propagator is exactly a Gaussian Fourier integral operator.

Probabilistic closure

Brownian motion is Gaussian and the Feynman–Kac weight is quadratic. The resulting functional integral is again Gaussian.

Ground-state closure

The logarithmic derivative of the Gaussian ground state is linear:

$$2\nabla \log \phi_0 = -2x.$$

Thus the ground-state transform produces a linear Gaussian diffusion—the Ornstein–Uhlenbeck process.

The different Riccati equations encountered in the kernel ansatz, the Weyl-symbol ansatz, the symplectic block decomposition, and the Gaussian covariance calculation are different coordinate expressions of the same linear Hamiltonian system.

12. A compact unified diagram

Let

$$H = -\Delta + |x|^2, \quad \phi_0 = e^{-|x|^2/2}, \quad \mathcal{L} = -\Delta + 2x \cdot \nabla.$$

Then

$$\boxed{\phi_0^{-1}(H - n)\phi_0 = \mathcal{L}}$$

and, for complex time,

$$\boxed{\phi_0^{-1}e^{-zH}\phi_0 = e^{-nz}e^{-z\mathcal{L}}.}$$

On the positive real axis,

$$z = t > 0 :$$

$$e^{-tH} \longleftrightarrow \begin{cases} \text{Mehler heat kernel,} \\ \text{Weyl Gaussian with } \tanh t, \\ \text{Feynman–Kac weighted Brownian motion,} \\ \text{OU Markov semigroup after the ground-state transform.} \end{cases}$$

On the imaginary axis,

$$z = it :$$

$$e^{-itH} \longleftrightarrow \begin{cases} \text{metaplectic quantization of } \Phi_t, \\ \text{oscillatory Mehler kernel,} \\ \text{exact Egorov evolution,} \\ \text{time-sliced Feynman oscillatory integral.} \end{cases}$$

The heat and unitary groups are therefore not two unrelated theories. They are two boundary regimes of the same Gaussian representation of the complexified symplectic dynamics.

Most relevant reading order in Taylor

1. **Chapter 7, §13, book pp. 56–67:** generic heat parametrix and transport equations.
2. **Chapter 7, §14, pp. 67–80:** Weyl calculus, Heisenberg group, symplectic covariance, and improved heat parametrix.
3. **Chapter 7, §15, pp. 80–88:** global oscillator symbol classes and the exact Weyl symbol of e^{-tH} .
4. **Chapter 8, §6, pp. 126–134:** ladder operators, spectrum, the $\mathfrak{sl}_2$ factorization, Mehler’s kernel, and general positive quadratic forms.
5. **Chapter 11, §§1–2, pp. 363–374:** Wiener measure and Feynman–Kac; especially the exercises on the quadratic potential.
6. **Chapter 11, §7, pp. 412–423:** stochastic-integral representations, Itô’s formula, and Langevin dynamics.
7. **Chapter 11, §§9–10, pp. 430–448:** SDEs and their Markov diffusion semigroups.
8. **Chapter 11, Appendix A, p. 448 onward:** the Trotter product formula underlying both time slicing and Feynman–Kac.

Could you make explicit the different appearances of Riccati equations (in the kernel ansatz, the Weyl-

symbol ansatz, the symplectic block decomposition, and the Gaussian covariance calculation) and explain how and why the entire complex semigroup $e^{\{-zH\}}$ ($\operatorname{Re} z \geq 0$) is dictated by the same linear Hamiltonian system?

Could you also fill Taylor's gaps, in the sense of extending his analysis of the heat-semigroup to the positive complex one?

1. The linear system behind everything

Use Taylor's normalization

$$H = -\Delta + |x|^2 = \operatorname{Op}^w(q), \quad q(x, \xi) = |x|^2 + |\xi|^2,$$

and set

$$J = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}.$$

The real classical Hamiltonian flow of q satisfies

$$\frac{d}{dt} \begin{pmatrix} x \\ \xi \end{pmatrix} = 2J \begin{pmatrix} x \\ \xi \end{pmatrix},$$

so

$$\Phi_t = e^{2tJ} = \begin{pmatrix} \cos 2t I & \sin 2t I \\ -\sin 2t I & \cos 2t I \end{pmatrix}.$$

To pass from the unitary group e^{-itH} to the heat operator e^{-zH} , put $t = -iz$. The relevant complex symplectic flow is therefore

$$S(z) = \Phi_{-iz} = e^{-2izJ} = \begin{pmatrix} \cosh 2z I & -i \sinh 2z I \\ i \sinh 2z I & \cosh 2z I \end{pmatrix}.$$

This matrix is entire in z and satisfies

$$S(z + w) = S(z)S(w).$$

The operator family is not entire as a bounded-operator-valued function: e^{-zH} is bounded precisely for $\Re z \geq 0$. The correct statement is that it is a **holomorphic semigroup on $\Re z > 0$** , strongly continuous on the closed half-plane.

Stable and unstable coordinates

Introduce the complex phase-space coordinates

$$u = x + i\xi, \quad v = x - i\xi.$$

The complex Hamilton equations become

$$u' = -2u, \quad v' = 2v,$$

and hence

$$u(z) = e^{-2z}u(0), \quad v(z) = e^{2z}v(0).$$

Let

$$\rho(z) = e^{-2z}.$$

Every coefficient in Mehler's formula, the Weyl symbol, the Gauss decomposition, and the Ornstein–Uhlenbeck covariance is a rational or Möbius function of this one solution of the linear equation

$$\rho' = -2\rho.$$

For example,

$$\begin{aligned}\tanh z &= \frac{1 - \rho}{1 + \rho}, \\ \tanh 2z &= \frac{1 - \rho^2}{1 + \rho^2}, \\ \coth 2z &= \frac{1 + \rho^2}{1 - \rho^2}, \\ \operatorname{csch} 2z &= \frac{2\rho}{1 - \rho^2}, \\ \frac{1 - e^{-4z}}{2} &= \frac{1 - \rho^2}{2}.\end{aligned}$$

This is the basic answer to why so many Riccati equations appear:

A Riccati equation is what a linear Hamiltonian system looks like after passing to a graph coordinate, a block ratio, an inverse covariance, or a Cayley coordinate.

Taylor identifies the relevant $\mathfrak{sl}_2$ -algebra and uses its Gauss factorization to motivate Mehler's formula, while noting that integrating the Lie-algebra representation requires additional analytic work.

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2. The general projectivization mechanism

Suppose a linear system is written in blocks as

$$\frac{d}{dz} \begin{pmatrix} X \\ Y \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix}.$$

Whenever X is invertible, the graph slope

$$P = YX^{-1}$$

satisfies

$$\boxed{P' = c + dP - Pa - PbP.}$$

The quadratic term PbP is the Riccati term. Nothing nonlinear has entered at the fundamental level: the nonlinearity appears because one divides one block of a linear solution by another.

For the complex oscillator system

$$S' = -2iJS,$$

we have

$$a = d = 0, \quad b = -2iI, \quad c = 2iI,$$

so

$$P' = 2i(I + P^2).$$

Writing $P = iA$ gives

$$\boxed{A' = 2(1 - A^2).}$$

That equation will reappear in several different coordinate systems.

3. Riccati equation from the kernel ansatz

For $\Re z > 0$, seek the kernel in the symmetric Gaussian form

$$K_z(x, y) = C(z) \exp \left[-\frac{1}{2} (A(z)(|x|^2 + |y|^2) - 2B(z)x \cdot y) \right].$$

Since

$$\partial_z e^{-zH} = -He^{-zH},$$

the kernel satisfies

$$\partial_z K_z = (\Delta_x - |x|^2) K_z.$$

A direct differentiation gives

$$\frac{\Delta_x K_z}{K_z} = -nA + A^2|x|^2 - 2ABx \cdot y + B^2|y|^2.$$

Comparison of coefficients yields

$$\begin{aligned} A' &= 2(1 - A^2), \\ B' &= -2AB, \\ \frac{C'}{C} &= -nA, \end{aligned}$$

together with

$$A' = -2B^2.$$

The last two versions of the equation for A' imply the first integral

$$A^2 - B^2 = 1.$$

This is the scalar shadow of the symplectic determinant condition.

The solution having the delta-function short-time behavior is

$$A(z) = \coth 2z, \quad B(z) = \operatorname{csch} 2z,$$

and

$$C(z) = (2\pi \sinh 2z)^{-n/2}.$$

Thus

$$K_z(x, y) = \frac{\exp \left[-\frac{\cosh 2z(|x|^2 + |y|^2) - 2x \cdot y}{2 \sinh 2z} \right]}{(2\pi \sinh 2z)^{n/2}}.$$

For positive real $z = t$, this is Taylor's Mehler kernel. He derives it by the Lie-algebra factorization and then verifies it directly as the heat solution.

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Linearization of the kernel Riccati equation

Let

$$s(z) = \sinh 2z.$$

Then

$$s'' = 4s, \quad A = \frac{1}{2} \frac{s'}{s}.$$

Consequently

$$A' = \frac{1}{2} \frac{s''}{s} - \frac{1}{2} \left(\frac{s'}{s} \right)^2 = 2 - 2A^2.$$

So the kernel Riccati equation is simply the logarithmic-derivative form of the imaginary-time Jacobi equation

$$s'' = 4s.$$

Relation to the symplectic blocks

Write

$$S(z) = \begin{pmatrix} a(z) & b(z) \\ c(z) & d(z) \end{pmatrix}.$$

For the oscillator,

$$b(z) = -i \sinh 2z I, \quad d(z) = \cosh 2z I.$$

The kernel coefficients are

$$\boxed{A = -i d b^{-1}, \quad B = -i b^{-1}.}$$

Indeed,

$$-i d b^{-1} = \coth 2z I, \quad -i b^{-1} = \operatorname{csch} 2z I.$$

The kernel Riccati equation is therefore exactly the Riccati equation satisfied by the graph of the propagated vertical Lagrangian

$$S(z)\{(0, \eta)\}.$$

4. Riccati equation from the Weyl-symbol ansatz

Taylor writes

$$e^{-tH} = \operatorname{Op}^w(h_t)$$

and uses the $U(n) \subset Sp(n, \mathbb{R})$ invariance of

$$Q(x, \xi) = |x|^2 + |\xi|^2$$

to justify the ansatz

$$h_z(x, \xi) = \alpha(z) e^{-\beta(z) Q(x, \xi)}.$$

Taylor's symmetry argument and the resulting real-time formula are in Chapter 7, §15.

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The underlying symplectic covariance of the Weyl product is developed in §14.

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Because Q is quadratic, the Moyal product terminates:

$$Q \# h = Qh + \frac{i}{2} \{Q, h\} - \frac{1}{4} \Delta_{x, \xi} h.$$

For a radial function $h = g(Q)$,

$$\{Q, h\} = 0.$$

Since phase space has dimension $2n$,

$$\Delta_{x, \xi} e^{-\beta Q} = (-4n\beta + 4\beta^2 Q) e^{-\beta Q}.$$

Therefore

$$Q \# h = [(1 - \beta^2)Q + n\beta] h.$$

The equation

$$\partial_z h_z = -Q \# h_z$$

becomes

$$\boxed{\begin{aligned} \beta' &= 1 - \beta^2, \\ \frac{\alpha'}{\alpha} &= -n\beta. \end{aligned}}$$

With $\beta(0) = 0$, $\alpha(0) = 1$,

$$\beta(z) = \tanh z, \quad \alpha(z) = (\cosh z)^{-n}.$$

Hence

$$\boxed{h_z(x, \xi) = (\cosh z)^{-n} e^{-(\tanh z)(|x|^2 + |\xi|^2)}.$$

Taylor obtains this for $t > 0$ and then recovers it from the integral kernel by a Gaussian Fourier transform.

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Why the argument contains z , rather than $2z$

The kernel uses a generating function associated with an off-diagonal block $b(z)$ of $S(z)$. The Weyl symbol instead uses the **Cayley transform**

$$\mathcal{C}(S) = (S - I)(S + I)^{-1}.$$

For the oscillator,

$$(S(z) - I)(S(z) + I)^{-1} = -i \tanh z J.$$

Consequently,

$$J(S - I)(S + I)^{-1} = i \tanh z I.$$

In the present normalization, the Weyl symbol can be expressed as

$$h_z(X) = \det \left(\frac{I + S(z)}{2} \right)^{-1/2} \exp [iX^T J(S(z) - I)(S(z) + I)^{-1} X],$$

where $X = (x, \xi)$.

Now

$$\det \left(\frac{I + S(z)}{2} \right)^{-1/2} = (\cosh z)^{-n},$$

and the exponential is

$$e^{-(\tanh z)|X|^2}.$$

Thus the Weyl Riccati coefficient is the Cayley coordinate of the same symplectic matrix. The half-angle is not accidental:

$$\tanh z = \tanh \left(\frac{1}{2} \log S(z) \right)$$

in matrix-functional-calculus language.

5. Riccati equation in Taylor's symplectic Gauss decomposition

After a complex change of basis, the same symplectic flow becomes

$$G(z) = \begin{pmatrix} \cosh 2z & \sinh 2z \\ \sinh 2z & \cosh 2z \end{pmatrix}.$$

It has the Gauss decomposition

$$G(z) = \begin{pmatrix} 1 & r(z) \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \operatorname{sech} 2z & 0 \\ 0 & \cosh 2z \end{pmatrix} \begin{pmatrix} 1 & 0 \\ r(z) & 1 \end{pmatrix},$$

where

$$r(z) = \tanh 2z.$$

Differentiation gives

$$r' = 2(1 - r^2), \quad r(0) = 0.$$

This is precisely the Riccati equation produced when one differentiates a triangular factorization of the linear matrix equation $G' = 2KG$.

Taylor's parameters satisfy, up to the sign convention used for the nilpotent generators,

$$\varepsilon_1(z) = \varepsilon_2(z) = \frac{1}{2} \tanh 2z.$$

Under the representation

$$X = -\partial_x^2, \quad Y = x^2, \quad Z = 2x\partial_x + 1,$$

the three matrix factors become:

- a free heat operator;
- a dilation;
- multiplication by a Gaussian.

Their composition gives the Mehler kernel. This is exactly Taylor's passage from equations (6.26)–(6.31) to the operator factorization (6.27).

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Notice the relationships

$$r = \tanh 2z, \quad A = \coth 2z = \frac{1}{r}.$$

The regular Gauss coordinate $r(0) = 0$ and the singular kernel coordinate $A \sim (2z)^{-1}$ are two reciprocal charts on the same evolving Lagrangian.

6. Riccati equations from Gaussian integration and covariance

There are two closely related probabilistic appearances.

6.1 Feynman-Kac Gaussian ansatz

Taylor proves

$$e^{-tH} f(x) = \mathbb{E}_x \left[f(\omega_t) \exp \left(- \int_0^t |\omega_s|^2 ds \right) \right]$$

by applying Feynman-Kac to the potential $V(x) = |x|^2$. His proof is based on the Trotter product formula. Michael Taylor - Partial Differ...

The exercises immediately following ask the reader to use Mehler's formula to evaluate the corresponding quadratic Brownian functional. Michael Taylor - Partial Differ...

Apply the semigroup to the constant function:

$$u(z, x) = e^{-zH} 1(x).$$

Use the Gaussian ansatz

$$u(z, x) = a(z) e^{-d(z)|x|^2}.$$

The PDE

$$\partial_z u = (\Delta - |x|^2)u$$

gives

$$d' = 1 - 4d^2, \quad \frac{a'}{a} = -2nd.$$

Therefore

$$d(z) = \frac{1}{2} \tanh 2z, \quad a(z) = (\cosh 2z)^{-n/2},$$

so

$$e^{-zH} 1(x) = (\cosh 2z)^{-n/2} \exp \left[-\frac{1}{2} \tanh 2z |x|^2 \right].$$

This coefficient is exactly half the Gauss parameter:

$$d = \frac{r}{2}.$$

Analytically, integrating the Mehler kernel in the endpoint y takes a Schur complement of its quadratic precision matrix. Riccati equations are stable under precisely such Gaussian eliminations.

6.2 Ornstein-Uhlenbeck covariance and its inverse

Taylor identifies the ground state

$$\phi_0(x) = \pi^{-n/4} e^{-|x|^2/2}, \quad H\phi_0 = n\phi_0,$$

as part of his ladder-operator treatment.

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With

$$Uf = \phi_0 f,$$

one calculates

$$U^{-1}(H - n)U = -\Delta + 2x \cdot \nabla.$$

Thus

$$e^{-zH} = e^{-nz} U e^{z(\Delta - 2x \cdot \nabla)} U^{-1}.$$

For real $t > 0$, the latter is the Ornstein-Uhlenbeck semigroup generated by

$$G = \Delta - 2x \cdot \nabla.$$

Its process is

$$dX_t = -2X_t dt + d\omega_t$$

in Taylor's convention $d\langle\omega\rangle_t = 2 dt$. Taylor discusses the exponentially damped Langevin dynamics and explicitly points toward Ornstein and Uhlenbeck, though he does not make this ground-state identification with H .

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The solution has mean

$$m_t = e^{-2t} x$$

and covariance

$$\Sigma_t = 2 \int_0^t e^{-4(t-s)} ds = \frac{1 - e^{-4t}}{2} I.$$

The covariance satisfies the linear Lyapunov equation

$$\Sigma' = 2I - 4\Sigma.$$

The inverse covariance, or precision matrix,

$$R = \Sigma^{-1},$$

satisfies

$$R' = -R\Sigma'R,$$

hence

$$R' = 4R - 2R^2.$$

Thus a linear covariance equation becomes a Riccati equation after inversion.

For complex z with $\Re z > 0$, define the analytic continuations

$$\Sigma(z) = \frac{1 - e^{-4z}}{2}I, \quad R(z) = \frac{2}{1 - e^{-4z}}I.$$

For nonreal z , $\Sigma(z)$ is no longer a probabilistic covariance, but it remains the complex Gaussian covariance of the analytically continued kernel.

The complex OU kernel is

$$p_z^{OU}(x, y) = \frac{\exp\left[-\frac{|y - e^{-2z}x|^2}{1 - e^{-4z}}\right]}{[\pi(1 - e^{-4z})]^{n/2}}.$$

The ground-state transform gives

$$K_z^H(x, y) = e^{-nz} \frac{\phi_0(x)}{\phi_0(y)} p_z^{OU}(x, y).$$

The coefficient dictionary is

$$\begin{aligned} R(z) &= 1 + \coth 2z, \\ e^{-2z} R(z) &= \operatorname{csch} 2z, \\ 1 + e^{-4z} R(z) &= \coth 2z. \end{aligned}$$

So the OU mean and covariance reconstruct exactly the two coefficients of the Mehler kernel.

7. One dictionary for all the Riccati equations

Let

$$\rho = e^{-2z}.$$

Appearance	Coefficient	Equation	Expression through ρ
Kernel diagonal Hessian	$A = \coth 2z$	$A' = 2(1 - A^2)$	$\frac{1 + \rho^2}{1 - \rho^2}$
Kernel cross coefficient	$B = \operatorname{csch} 2z$	$B' = -2AB$	$\frac{2\rho}{1 - \rho^2}$
Weyl Gaussian exponent	$\beta = \tanh z$	$\beta' = 1 - \beta^2$	$\frac{1 - \rho}{1 + \rho}$
Gauss shear	$r = \tanh 2z$	$r' = 2(1 - r^2)$	$\frac{1 - \rho^2}{1 + \rho^2}$
Feynman-Kac endpoint- integrated Gaussian	$d = \frac{1}{2} \tanh 2z$	$d' = 1 - 4d^2$	$\frac{1 - \rho^2}{2(1 + \rho^2)}$
OU covariance	$\Sigma = \frac{1}{2}(1 - \rho^2)$	$\Sigma' = 2 - 4\Sigma$	$\frac{1 - \rho^2}{2}$
OU precision	$R = \Sigma^{-1}$	$R' = 4R - 2R^2$	$\frac{2}{1 - \rho^2}$

The sole primitive dynamical quantity here is

$$\rho' = -2\rho.$$

The different Riccati equations come from applying:

- a block ratio;
- a reciprocal;
- a Cayley transform;
- a Schur complement;
- or matrix inversion

to the same linear flow.

8. Extending Taylor's formulas to the full right half-plane

Taylor establishes the formulas for real $t > 0$. The extension to $\Re z > 0$ can be made rigorous with relatively little additional work.

8.1 Spectral definition

Let h_α be the Hermite basis, so

$$Hh_\alpha = (2|\alpha| + n)h_\alpha.$$

Define

$$e^{-zH}f = \sum_{\alpha \in \mathbb{N}^n} e^{-z(2|\alpha|+n)} \langle f, h_\alpha \rangle h_\alpha, \quad \Re z \geq 0.$$

For $\Re z > 0$, this is holomorphic as a bounded-operator-valued function and

$$\|e^{-zH}\|_{L^2 \rightarrow L^2} = e^{-n\Re z}.$$

Moreover,

$$e^{-zH}e^{-wH} = e^{-(z+w)H}$$

whenever $\Re z, \Re w \geq 0$.

For $\Re z > 0$, it is trace class, with

$$\mathrm{Tr}(e^{-zH}) = \sum_{\alpha} e^{-z(2|\alpha|+n)} = (2 \sinh z)^{-n}.$$

8.2 Holomorphic continuation of the Weyl formula

In the open half-plane $\Re z > 0$, neither $\cosh z$ nor $\sinh 2z$ vanishes. Therefore

$$h_z(x, \xi) = (\cosh z)^{-n} e^{-(\tanh z)(|x|^2 + |\xi|^2)}$$

is holomorphic in z .

For real $z = t > 0$, Taylor proves

$$e^{-tH} = \mathrm{Op}^w(h_t).$$

Both sides are holomorphic operator-valued functions on $\Re z > 0$. Since they agree on the positive real axis, the identity theorem gives

$$e^{-zH} = \mathrm{Op}^w(h_z), \quad \Re z > 0.$$

This is the cleanest way to fill Taylor's complex-time gap.

8.3 Holomorphic continuation of the kernel

Choose the branch of

$$(\sinh 2z)^{n/2}$$

that is positive for positive real z . Then

$$K_z(x, y) = \frac{\exp \left[-\frac{\cosh 2z(|x|^2 + |y|^2) - 2x \cdot y}{2 \sinh 2z} \right]}{(2\pi \sinh 2z)^{n/2}}, \quad \Re z > 0.$$

The exponent can be rewritten as

$$-\frac{1}{4} \tanh z |x + y|^2 - \frac{1}{4} \coth z |x - y|^2.$$

For $\Re z > 0$,

$$\Re \tanh z > 0, \quad \Re \coth z > 0,$$

so this is a genuinely decaying complex Gaussian. Thus K_z is a Schwartz kernel for every z in the open right half-plane.

Again, equality for positive real z , followed by holomorphic continuation, proves that this is the kernel of e^{-zH} .

9. Why the semigroup law is already contained in the Hamiltonian flow

At the symplectic level,

$$S(z + w) = S(z)S(w).$$

At the Weyl-symbol level, the Cayley parameter satisfies

$$\beta(z + w) = \frac{\beta(z) + \beta(w)}{1 + \beta(z)\beta(w)}, \quad \beta(z) = \tanh z.$$

The prefactor satisfies

$$(\cosh(z + w))^{-n} = (\cosh z)^{-n} (\cosh w)^{-n} (1 + \beta(z)\beta(w))^{-n}.$$

These are exactly the Gaussian Weyl-product composition rules

$$h_z \# h_w = h_{z+w}.$$

At the kernel level, Gaussian convolution implements block multiplication of $S(z)$ and $S(w)$; the hyperbolic addition formulas are simply the scalar form of that block multiplication.

At the probabilistic level, the OU formulas give

$$e^{-2(z+w)} = e^{-2z} e^{-2w}$$

and

$$\Sigma(z+w) = \Sigma(w) + e^{-4w} \Sigma(z),$$

which is the covariance form of the Chapman-Kolmogorov identity.

Thus the four semigroup laws are the same identity viewed in:

- symplectic matrices;
- Weyl symbols;
- integral kernels;
- Gaussian transition laws.

10. Boundary values: the unitary group and caustics

For every $f \in L^2$,

$$\lim_{\varepsilon \downarrow 0} e^{-(\varepsilon+it)H} f = e^{-itH} f$$

strongly in L^2 .

This follows immediately from the spectral expansion:

$$e^{-(\varepsilon+it)(2|\alpha|+n)} \longrightarrow e^{-it(2|\alpha|+n)}.$$

When

$$\sin 2t \neq 0,$$

the Mehler kernel has the oscillatory boundary value

$$K_{it}(x, y) = \frac{\exp \left[i \frac{\cos 2t(|x|^2 + |y|^2) - 2x \cdot y}{2 \sin 2t} \right]}{(2\pi i \sin 2t)^{n/2}},$$

with the square-root phase selected by approaching from $\Re z > 0$.

This is the metaplectic quantization of

$$\Phi_t = e^{2tJ}.$$

At the caustic times

$$t \in \frac{\pi}{2}\mathbb{Z},$$

the kernel formula ceases to be an ordinary function because the block $b(it)$ is singular. The operator itself remains perfectly well defined.

For $k \in \mathbb{Z}$,

$$e^{-ik\pi H} = e^{-ik\pi n} I,$$

while

$$e^{-i(2k+1)\pi H/2} = e^{-i(2k+1)\pi n/2} \mathcal{P}, \quad (\mathcal{P}f)(x) = f(-x).$$

Similarly, the Weyl formula

$$h_{it}(x, \xi) = (\cos t)^{-n} e^{-i(\tan t)(|x|^2 + |\xi|^2)}$$

is an ordinary symbol whenever $\cos t \neq 0$. At $\cos t = 0$, that Weyl chart becomes singular and the symbol has to be understood distributionally.

These singularities are chart singularities, not singularities of the unitary group. They are the metaplectic version of caustics.

11. Why $\Re z > 0$ is the natural semigroup domain

The complex symplectic matrix $S(z)$ exists for every $z \in \mathbb{C}$. But the operator e^{-zH} is bounded only when $\Re z \geq 0$.

In stable coordinates,

$$\rho = e^{-2z}.$$

- If $\Re z > 0$, then $|\rho| < 1$. The stable mode contracts, the covariance $1 - \rho^2$ is nonzero, and the Gaussian kernel has positive real quadratic part.
- If $\Re z = 0$, then $|\rho| = 1$. Decay is lost and the kernel becomes oscillatory; at isolated points the chosen Gaussian chart develops a caustic.

- If $\Re z < 0$, then $|\rho| > 1$, and

$$|e^{-z(2|\alpha|+n)}| = e^{-\Re z(2|\alpha|+n)}$$

grows without bound as $|\alpha| \rightarrow \infty$.

Thus the same linear Hamiltonian flow exists globally in complex time, but its positive/stable region is exactly the right half-plane.

12. General positive quadratic Hamiltonians

Taylor already indicates how to pass from the isotropic oscillator to an arbitrary positive quadratic form $Q(x, \xi)$: symplectically diagonalize Q , introduce its Hamilton map, and write the result invariantly.

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In general, let

$$q(X) = \frac{1}{2} X^T G X$$

and let $F = JG$ be the Hamilton matrix. The complexified linear flow is

$$S_q(z) = e^{-izF}.$$

Then:

- the kernel phase is obtained from block ratios such as DB^{-1} ;
- the Weyl quadratic form is obtained from

$$J(S_q - I)(S_q + I)^{-1};$$

- triangular factor coordinates satisfy matrix Riccati equations;
- Gaussian covariances satisfy Lyapunov equations, whose inverses satisfy matrix Riccati equations.

So the harmonic oscillator is not merely one example where several calculations happen to produce hyperbolic functions. It is the simplest case of a general principle:

Quadratic quantum evolution is the representation-theoretic image of a linear

The Riccati equations are the various projective coordinate systems in which that linear flow is observed.

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